

A Comprehensive Study of Spatiotemporal Feature Learning for Social Media Popularity Prediction

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ABSTRACT

For accurately predicting the popularity of social media, the multi-modal approach was usually adopted to have promising performance. However, the popularity is highly correlated to its identity (i.e., user ID). Inappropriate data splitting could result in lower generalizability in real-world applications. Specifically, we observed that the training and testing datasets are partitioned on a specific timestamp, whereas some users were registered after the timestamp, implying that the partial identities could be missing in the testing phase. It turns the social media prediction (SMP) tasks temporally irrelevant, making the temporal-related feature useless due to missing identities. Therefore, we form the SMP task as an identity-preserving time-series task to observe the popularity scores for the specific identity. In addition, more valuable and essential features could be explored. In this paper, by reformulating the SMP tasks and integrating the multi-modal feature aggregation to the base learner for better performance, how the identity-preserving is an essential property for SMP tasks is discussed. We firstly explore the impact of the temporal features with/without identity information in the conventional SMP tasks. Moreover, we reformulate the SMP task in the time-series data-splitting and evaluate the temporal features' importance. Comprehensive experiments are conducted to deliver the suggestions for the SMP tasks and offer the corresponding solutions for effectively predicting the popularity scores.

CCS CONCEPTS

• **Computing methodologies** → **Ensemble methods**; *Feature selection*;

KEYWORDS

Regression; multi-modal, time-series, identity-preserving, context.

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1 INTRODUCTION

In recent years, the virtual world has gone hand in hand with the natural world and has become a part of our life. People share their daily routine on social media and do care about whether their posts are famous or not. It drives that the popularity prediction of social media posts has become a popular research topic in recent years. Many studies are committed to precisely predicting the popularity score of a post. It is well-known that the data type of social media was highly variant, such as images, photo categories, text information, geographic data, spatiotemporal feature, and metadata. How to effectively deal with the heterogeneous data adequately to fit the model is essential.

In traditional SMP task [1–3], the data was partitioned into the training and testing sets by the timestamp only. However, some users in the testing set would be different from that in the training set. In [8–10], the user identity (UID) is the most crucial feature for SMP tasks. Therefore, inappropriate data splitting could even restrict the performance of the SMP task due to missing UIDs (i.e., identity). As for accurately predicting the popularity of social media, the multi-modal approach was usually adopted to have promising performance. However, the popularity is highly correlated to its identity (i.e., user ID). Inappropriate data splitting could result in the suppressed performance of the social media prediction. Specifically, we observed that the training and testing datasets are partitioned on a specific timestamp, whereas partial users were registered after the timestamp, implying that the partisan identities could be missing in the testing phase. It turns the social media prediction (SMP) tasks temporally irrelevant, making the temporal-consistency feature ineffective due to missing identities. We argue that the SMP could be formed as an identity-aware time-series task

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since we could observe the changes in the popularity scores in time for a specific identity. Furthermore, it allows us to explore more valuable and essential features in the temporal domain rather than only identity-related features, making the model even more robust and effective. In this paper, we fully discover the effectiveness and influence of the temporal features in SMP tasks by reformulating them to be identity-preserving time-series ones, as well as integrating the multi-modal feature aggregation to the base learner for better performance.

This paper fully explores the variants of conventional SMP task [1–3] by involving the proposed identity-preserving time-series (IT) data splitting strategy to reformulate the SMP, termed IT-SMP task. Furthermore, more essential and valuable temporal and traditional time-series models could be used in our IT-SMP task to boost performance further.

The prior art [8–10] SMP models treat the prediction task as the multi-modal regression task. In the feature extraction, the Word-to-Vector (W2V) model was widely adopted to encode the frequency of words in the text information. Term Frequency-Inverse Document Frequency (TF-IDF) model has also been proposed in [12] to decrease the influence of the insignificant high-frequency vocabulary in the text. As for the semantic features of the text information, the conventional nature-language-process (NLP) models are usually adopted, such as RoBERTa [14]. In [8] claims that the image features and its generative caption could even improve the performance of the SMP task. However, rare studies focus on investigating the essential temporal feature since the conventional SMP task lacks temporal consistency for each identity.

With the extracted features from social media, unsurprisingly, the traditional discrete regression schemes, such as XGBoost (XGB) [4], Random Forest (RF) [13], Light-GBM (LGBM) [11], CatBoost (CATB) [17], and even model stacking [16] were adopted to deal with the popularity prediction. Supervised learning and ensemble-based strategies are superior to various tasks, especially in tabular data. In this paper, a multi-modal feature aggregation strategy is proposed and followed by adopting the discrete regression model to explore the excellent performance as possible.

The second type dealing with the SMP task is the deep learning-based approach, which performs the regression or classification tasks excellently by joint learning the feature representation and classifier in the nonlinear domain. However, the single deep learning-based model appears not superior to the ensemble learning-based model by comparing the performance indicators [5]. However, in [10], the authors claim that the model stacking problem could be turned into a recurrent neural network (RNN) and proposed an end-to-end trainable model stacking, termed the Recurrent Stacking Model (RSM). However, as we mentioned previously, all the conventional SMP schemes fail to discuss the time-series-type regression due to the missing identity issue. The generalizability could be even worse based on the traditional SMP tasks since the identity might be missing while the identity-related features dominate the SMP performance.

As for tickling the time-series prediction issue, the hard condition of the data splitting should be identity-preserving property. With the variant version of the traditional SMP task (i.e., IT-SMP), we turn the popularity prediction from discrete regression into time-series regression. Since the period of the posts for a specific

user could be even longer, the traditional Long Short-term Memory Network (LSTM) [6][7][18] and its variants could fail to capture the long-range correlation. Transformer [19] is a well-known scheme that can capture the long-range dependency by the self-attention mechanism. However, the computational complexity could be high as the period of the posts is too long. In this paper, we also investigate the efficient self-attention proposed by Informer [22], probability-sparse self-attention (PSSA), to reduce complexity without performance degradation significantly.

In summary, a novel discrete regression based on multi-modal feature aggregation as well as the new IT-SMP task is jointly investigated to fully explore the impact of the temporal features on time-series prediction. First, we discuss the impact of the temporal features with/without identity information in the conventional social media prediction task. Second, the task is reformulated in the time-series data splitting. Comprehensive experiments are conducted to deliver the suggestions for improving the current SMP task, as well as the PSSA is proposed to capture the temporal correlation in the IT-SMP to achieve state-of-the-art performance. The contribution of this paper is two-fold:

- We fully discover the possible variant of conventional SMP task to form IT-SMP task in time-series view, as well as propose a novel Transformer-based regression to verify the importance of the temporal features.
- A novel multi-modal feature aggregation module is also proposed for conventional SMP based on the discrete regression scheme.

The rest of this paper is organized as follows. Section 2 presents the proposed multi-modal feature aggregation module for the SMP task. In Section 3, we show that the proposed IT-SMP task could be even more challenging as well as more temporal features could be useful. Section 4 shows our experimental results. Finally, conclusions are drawn in Section 5.

2 MULTI-MODAL FEATURE AGGREGATION

We firstly drive the conventional SMP task based on the proposed multi-modal feature aggregation module. There are four types of SMP data: 1) numerical, 2) categorical, 3) image-based, and 4) text-based data. Most of the preprocessing techniques for those four types of SMP data are referred to as [8]. This study proposes an enhanced feature representation for text-based data to retrieve more semantic information. The conventional regression schemes, such as LGBM, XGB, and RF, could be used on the aggregated multi-modal features to learn the regressor for SMP tasks.

2.1 Data Preprocessing and Analysis

The naive social media post information we adopted can be organized into a 15-point feature, including numerical, categorical, text-based information, and an image. Most of the data parsing schemes are followed the suggestions in [10].

Image Data Analysis. As suggested in [8], the image captioning model trained on MS-COCO is used to extract the caption from images, followed by encoding the caption using the W2V model to extract the semantic feature f_{cap} . Meanwhile, the image feature from ResNet-50 is extracted and followed by reducing its dimensionality to form the compact image feature f_i .

Numerical Data Analysis. We transfer the UID and post-ID (PID) into the unique integer. As for Postdate, we convert the publish timestamp of the post to Datetime, which is indicated by f_t . The rest of the numerical data is denoted as f_n .

Text-based Data Analysis. Inspired by [8][10][12], we choose each of the text-based data to train individual W2V, RoBERTa, and Term Frequency-Inverse Document Frequency (TF-IDF) models, denoted by f_{w2v} , f_r , and f_f . Since the dimensionality of the f_r and f_f could be high if the number of the words is large, the principal component analysis (PCA) is adopted to reduce their dimensionality.

Categorical Data Analysis. For the three categorical data, we encode them as a one-hot vector representation, denoted as f_c .

Feature Aggregation. After processing all the usable information, we aggregated the multi-modal features $f_{all} = [f_{cap}, f_i, f_t, f_n, f_{w2v}, f_f, f_r, f_c]$ as the reference of the social media popularity prediction.

2.2 k -fold Validation and Prediction

Directly training the regressor based on all training sets without validation supervision could cause overfitting. Therefore, k -fold cross-validation is used to improve the stability and prevent overfitting. For each fold, the partial training samples are used to train the regressor, as well as evaluated in the rest part for finding the best model. The regressor at each fold could be adopted to predict the popularity of the testing set, and the final result would be made by ensembling all of the predicted popularity scores.

3 TIME-SERIES-BASED PREDICTION

As we aforementioned, the conventional SMP task could be identity inconsistent between training and testing phases, leading to suppressed performance. Specifically, the numbers of users of the training and that of the private testing set (no label accessible) are 38,312 and 31,392, respectively. 21,781 users in the private testing set cannot be found on the training set, approximating 69%. In other words, the conventional SMP task is missing the identity-aware temporal consistency, causing performance degradation. We turn the conventional SMP to the time-series-based arrangement and form the IT-SMP task to discover more temporal features that benefit the popularity of the social post.

To keep the identity consistent between the training and testing set, we resample the original data based on the timestamp with identity consistency. The number of the historical posts less than t_h will be removed to avoid the insufficient number of historical data after data splitting. In this IT-SMP task, the time-series regression schemes could be used to predict the popularity of social posts. The proposed model in this section is shown in Fig. 1.

3.1 Time-series Regression

The time-series prediction task is usually tickled by the recurrent-like neural networks [20][22]. However, the lack of the ability to capture the long-range dependency at the temporal axis makes the performance suppressed on time-series prediction. Recently, Transformer [19] was proposed to resolve this issue based on the self-attention mechanism (SAM). However, the longer the sequence, the higher the computational complexity. Since the number of posts per identity could be tremendous, the complexity of the Transformer would be even higher. The probability-sparse self-attention

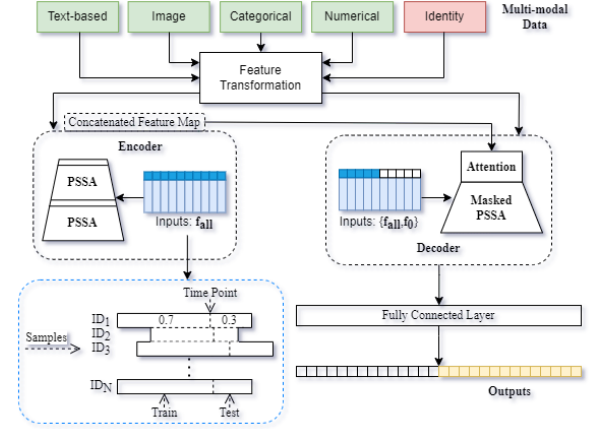


Figure 1: Flowchart of the proposed regression model with multi-modal feature mining.

(PSSA) is adopted to reduce the complexity, as shown in Fig. 1. Traditional SAM could be written as

$$A(q_i, K, V) = \sum_j \frac{k(q_i, k_j)}{\sum_l k(q_i, k_l)} v_j = \mathbb{E}_{p(k_j|q_i)}[v_j], \quad (1)$$

where $Q = \mathbb{R}^{L_Q \times d}$, $K = \mathbb{R}^{L_K \times d}$, $V = \mathbb{R}^{L_V \times d}$, d is the input dimension, and q_i, k_i , and v_i represent i -th row of query, key, and value vectors. The computational bottleneck is on computing $\mathbb{E}_{p(k_j|q_i)}$. However, inspired by Informer [22], the attention is usually sparse, making the complexity reduction possible. First, the KL-divergence is adopted to calculate the difference between p and q , where q is the uniform distribution, and p is the probability of being attentive. Therefore, the PSSA would be formulated as

$$A(Q, K, V) = \text{Softmax}\left(\frac{\bar{Q}K^T}{\sqrt{d}}\right)V. \quad (2)$$

To fully discover the power of the temporal feature for our IT-SMP task, we simply add the time-lag feature t_l calculated by the difference value between the post dates of two successive posts. In this way, we expect the temporal-related features to become the essential ones in our IT-SMP tasks. Finally, the smooth ℓ_1 -norm loss function is used to train the proposed PSSA with attention masking mechanism (masked PSSA) for our IT-SMP task.

4 EXPERIMENTAL RESULTS

The SMP Dataset (SMPD) provided in [1–3] consists of 305,614 posts with various and diverse data types. The handling of missing data is based on the way suggested in [10]. The evaluation metrics for performance evaluation are Spearman's Rho rank correlation (SRC)[21], Mean Absolute Error (MAE), and Mean Squared Error (MSE).

4.1 Evaluation of SMP Task

This subsection presents the performance evaluation of the traditional SMP task with the proposed multi-modal feature aggregation module. A single LightGBM model with the same hyperparameters to better observe the influence of the data processing

Table 1: Performance comparison among the different data processing methods evaluated on the validation set.

Methods	Data split	SRC	MSE	MAE
LGBM(w/o TF-IDF)[8]	identity-isolated	0.680	3.223	1.399
LGBM(w/ TF-IDF)	identity-isolated	0.680	3.201	1.395
k -fold LGBM($k=5$)	random	0.916	1.037	0.608

Table 2: Feature importance versus feature correlation.

	Importance	UID-Correlation
top1	UID	1
top2	PID	0.652
top3	Latitude	0.652
top4	Longitude	0.652
top5	Path-Alias	n/a
top6	Post Date	0.210

method. Two data splitting strategies are adopted to verify the impact of the identity-aware features. The first is the random sampling from the SMPD to form training and validation sets, while another will partition the SMPD into two identity-isolated sets.

The performance of the proposed multi-modal feature aggregation module based on random split and the identity-isolated split is drawn in Table 1. As we can see, the performance on the random split is significantly better than that on identity-isolated, implying that the identity-related features should be the critical clues for SMP tasks. Table 2 shows that the identity-related features indeed exist in the SMPD, leading to significant performance degradation when the identity-isolated split is adopted. Table 3 also shows that the performance degradation is more severe in the identity-isolation split than in the random split. It verifies that the identity-related features are essential for SMP tasks. However, in real-world applications, the number of identities could increase daily, implying that we should seek more essential features other than identity-related ones. It also implies that there exist several features that could leak UID-related information. Specifically, the feature correlation is conducted to verify the features that are highly correlated to UID, which exactly match the results of the feature importance. Essential features of the current SMP task would be identity-related features, making it hard to generalize. Rethinking the new SMP task, i.e., the proposed IT-SMP, could allow us to explore more potential and essential features in the spatial and temporal domain.

4.2 Evaluation of our IT-SMP Task

As for performance evaluation of our IT-SMP task, an identity-preserving random split on the temporal axis is adopted to reformulate the original SMPD. We follow the splitting ratio suggested in [10] to partition the SMPD into training and testing sets with a ratio of 8:2 on the temporal axis (i.e., post date). To ensure identity consistence, we remove the identity if its number of posts per identity is less than t_h , where $t_h = 10$ in this study.

The performance comparison between the proposed PSSA-based Transformer and conventional AutoRegressive Modeling (ARM)

Table 3: Performance of the proposed model based on random split and the identity-isolated split.

Methods	Uid	Pid	SRC	MSE	MAE
random split	✓	✓	0.891	1.320	0.738
random split		✓	0.888	1.356	0.748
random split			0.888	1.352	0.748
identity-isolated split	✓	✓	0.680	3.329	1.439
identity-isolated split		✓	0.666	3.469	1.472
identity-isolated split			0.662	3.467	1.473

Table 4: Performance evaluation of the proposed PSSA for IT-SMP task.

Methods	SRC	MAE
PSSA-1	0.785	1.117
PSSA-2	0.799	1.049
ARM	0.483	6.874

Table 5: Feature importance of IT-SMP task.

	Top1	Top2	Top3	Top4	Top5
PSSA-1	P_w	P_c	Title	f_{cap}	f_{im}
PSSA-2	t_l	P_w	f_{im}	f_{cap}	P_w

[20] in the Table 4. The proposed PSSA significantly outperforms the conventional ARM scheme. Moreover, the feature importance of the time-series setting, which is measured by a method in [15], is significantly different from that of conventional SMP tasks (see Table 5), as well as the identity-related features are no longer necessary. It also enables us to discover more temporal/spatial features for real-world applications. Note that the P_w and P_c denote the week of the post and the averaged interval between posts, and t_l is the time-lag feature discussed in Section 3.1. PSSA-1 and PSSA-2 are the proposed time-series predictor with/without time-lag feature t_l . It is also remarkable that the image feature seems to become more important in the IT-SMP tasks.

5 CONCLUSIONS

In this paper, we have proposed a multi-modal feature aggregation (MDFA) module and reformulated the social media prediction (SMP) task to the identity-preserving time-series one, termed IT-SMP. The proposed MDFA with the k -fold cross-validation could lead to performance improvement for conventional SMP tasks. In the proposed IT-SMP task, the probability-sparse self-attention mechanism (PSSA) with Transformer is proposed to capture the long-range dependency to achieve excellent performance. Finally, comprehensive studies have been conducted that the IT-SMP task could be more generalized, as well as remaining more room to discover temporal-related features for real-world SMP issues.

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